FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

CS0065
(INTELLIGENT SYSTEMS)

TECHNICAL ASSESSMENT

2

RULE – BASED AGENT

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I. BACKGROUND INFORMATION

A Rule-Based Agent is a type of intelligent agent that operates by following predefined rules to decide its actions. In this lab, students will work with a vacuum cleaner agent that can clean rooms or move between rooms depending on their state (dirty or clean).

The goal is to understand how agents perceive the environment, act according to rules, and update the environment.

II. TECHNICAL ASSESSMENT

RULE – BASED AGENT

Objective

At the end of this laboratory activity, students are expected to:

1. Explain the concept of rule-based agents.
2. Implement a rule-based agent in Python.
3. Interact with an environment by perceiving and acting based on rules.
4. Observe and analyze the behavior of an agent in a controlled environment.

Materials Needed

- Computer with Python 3.x installed
- Text editor or IDE (e.g., VSCode, PyCharm)
- Optional: Jupyter Notebook for interactive testing

General Instructions

1. This **Technical Assessment (TA4)** must be completed individually.
2. Follow the Task below.
3. Include clear comments in your code to describe its logic.
4. Each program should display the correct input, process, and output.
5. Ensure that your script displays clear and readable outputs.
6. Compile all Python codes and corresponding outputs into one PDF file.
7. Save it as **TA2_lastname.pdf**
   (Example: TA2_Patilano.pdf)
8. Submit the **python file** and **PDF** file.

Laboratory Task

Pre-Lab Preparation

1. Review the Python concepts:
   - Classes and objects

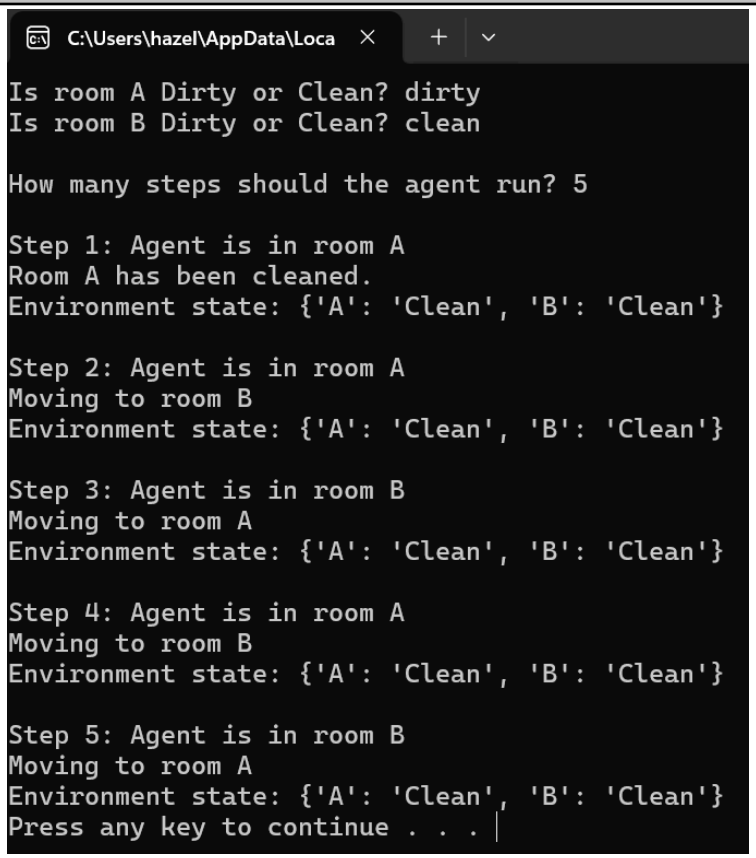
- If-Else statements
- Loops (for loop)
- Functions / methods

2. Understand the environment:
   - Two rooms: A and B
   - Each room can be Dirty or Clean

Task 1: Set Up the Environment

1. Initialize two rooms using a Python dictionary.
2. Allow the user to set the **initial state** of each room (dirty or clean).
3. Write a function to **check if a room is dirty** and a function to **clean a room**.

Sample Output



Task 2: Implement the Rule-Based Agent

1. Create a class for the agent.
2. Implement the **perceive-and-act method**:
   - Rule 1: If the current room is dirty, clean it.
   - Rule 2: If the current room is clean, move to the other room.
3. Implement a **move method** to switch rooms.

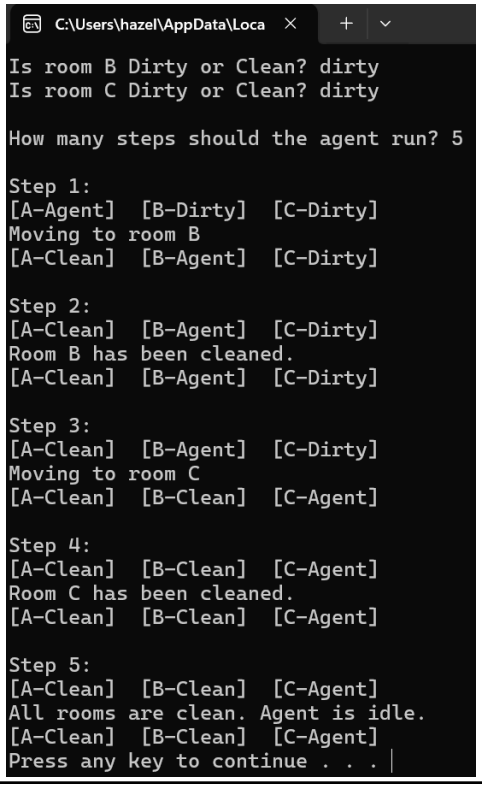
Task 3: Run the Simulation

1. Ask the user how many steps the simulation should run.
2. Run the agent in a loop for the specified steps.
3. After each step, print:
    - The agent's current location
    - The action taken (cleaned/moved)
    - Current state of the environment

Optional (Bonus Task)

- Add a **third room (C)** and update the agent logic.
- Make the agent **randomly choose which dirty room to clean next**.
- Visualize the agent moving between rooms using a **text-based grid**.

Sample Output (Bonus Task)



Your Code - Laboratory Tasks ⇩

```
Enter the initial state of Room A (Dirty/Clean): Clean
Enter the initial state of Room B (Dirty/Clean): Dirty
How many simulation steps should the agent run? 5

--- Rule-Based Agent Simulation ---
Initial environment: Room A = Clean, Room B = Dirty

Step 1
Agent location: Room B
Action taken: Moved from Room A to Room B
Environment: Room A = Clean, Room B = Dirty

Step 2
Agent location: Room B
Action taken: Cleaned Room B
Environment: Room A = Clean, Room B = Clean

Step 3
Agent location: Room A
Action taken: Moved from Room B to Room A
Environment: Room A = Clean, Room B = Clean

Step 4
Agent location: Room B
Action taken: Moved from Room A to Room B
Environment: Room A = Clean, Room B = Clean

Step 5
Agent location: Room A
Action taken: Moved from Room B to Room A
Environment: Room A = Clean, Room B = Clean
(.venv) asher@Ashers-MacBook-Air Python %
```

```python
def get_room_state(room_name):
    """Ask the user whether a room starts dirty or clean."""
    while True:
        state = input(
            f"Enter the initial state of Room {room_name} (Dirty/Clean): "
        ).strip().lower()

        if state in ("dirty", "clean"):
            return state.title()

        print("Invalid input. Please enter Dirty or Clean.")


def is_dirty(environment, room):
    """Check whether a room is dirty."""
    return environment[room] == "Dirty"


def clean_room(environment, room):
```

```
    """Clean a room."""
    environment[room] = "Clean"


class RuleBasedAgent:
    """An agent that cleans a dirty room or moves from a clean room."""

    def __init__(self, starting_room="A"):
        self.location = starting_room

    def move(self):
        """Switch between Rooms A and B."""
        self.location = "B" if self.location == "A" else "A"

    def perceive_and_act(self, environment):
        """Check the current room and apply one rule."""
        if is_dirty(environment, self.location):
            clean_room(environment, self.location)
            return f"Cleaned Room {self.location}"

        previous_room = self.location
        self.move()
        return f"Moved from Room {previous_room} to Room {self.location}"


def get_steps():
    """Ask for a positive number of simulation steps."""
    while True:
        try:
            steps = int(input("How many simulation steps should the agent run? "))
            if steps > 0:
                return steps
            print("Please enter a number greater than zero.")
        except ValueError:
            print("Invalid input. Please enter a whole number.")


def main():
    # Set up the two-room environment.
```

```
    environment = {
        "A": get_room_state("A"),
        "B": get_room_state("B"),
    }

    agent = RuleBasedAgent()
    steps = get_steps()

    print("\n--- Rule-Based Agent Simulation ---")
    print(
        f"Initial environment: Room A = {environment['A']}, "
        f"Room B = {environment['B']}"
    )

    # Run one perceive-and-act cycle per step.
    for step in range(1, steps + 1):
        action = agent.perceive_and_act(environment)

        print(f"\nStep {step}")
        print(f"Agent location: Room {agent.location}")
        print(f"Action taken: {action}")
        print(
            f"Environment: Room A = {environment['A']}, "
            f"Room B = {environment['B']}"
        )


if __name__ == "__main__":
    main()
```

Task 4: Analysis Questions

After running the simulation, answer the following:

1. What rules does the agent follow?

2. How does the agent decide when to clean and when to move?
3. What happens if both rooms are already clean at the start?
4. How would you modify the agent to handle **more than two rooms**?

Answer the Questions ⇩

1. The agent follows two rules: clean the current room if it is dirty; move to the other room if the current room is clean.
2. At each step, it checks the state of the room it is in. If that room is Dirty, it changes it to Clean and stays there. If it is already Clean, it moves.
3. It never cleans. It moves back and forth between Rooms A and B for the number of steps the user entered, because every room it checks is clean.
4. Store the room names in a list and track the agent's position in that list. To move, advance to the next position and wrap around to the first room after the last one.